

JOHN ABBOUD, C.E.T.

GIS Technologist in Training - Bachelor of Applied Technology, Geographic Information Systems, SAIT

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john-abboud.github.io

GIS Student currently completing SAIT's Bachelor of Applied Technology in Geographic Information Systems (BGIS), with a background in electronics spanning 15+ years across aerospace, oilfield, and utility industries. Experienced in composite spatial index development, PostgreSQL/PostGIS database implementation, spatial statistics, and enterprise/web GIS production using ArcGIS Pro and ArcGIS Online. Proven ability to collaborate with multidisciplinary teams, translate project requirements into GIS deliverables, and communicate complex spatial information to both technical and non-technical audiences. Brings a C.E.T. designation and a strong foundation in precision troubleshooting, root-cause analysis, and technical documentation that carries directly into GIS data quality and field work.

GIS EXPERIENCE

GIS Analyst - Student (Semester 1 Final Project) | SAIT, Calgary (Jan 2026 - Apr 2026)

Responsibilities

- Co-developed "Where the Sun Shines and Income Aligns," a GIS-based rooftop solar suitability analysis for Calgary, integrating solar radiation, residential rooftop area, and median household income into a composite suitability index.
- Built a solar radiation raster surface from LiDAR-derived DEM data and applied zonal statistics, spatial joins, and Esri Enrich demographic data at the community level.
- Designed and implemented a parallel PostgreSQL/PostGIS spatial database with an Entity Relationship Diagram, foreign key constraints, and advanced spatial SQL queries.
- Conducted spatial statistical analysis, including Hot Spot Analysis (Getis-Ord Gi*), Cluster and Outlier Analysis (Local Moran's I), and Global Moran's I, and produced bivariate choropleth mapping of solar potential against income.
- Automated the end-to-end workflow using ArcGIS ModelBuilder, from data cleaning and topology validation through composite index calculation.

Achievements

- Identified and ranked the top Calgary communities for rooftop solar development (Edgemont, Tuscany, Springbank Hill, Aspen Woods, among others), validated across two independent methods (composite index and bivariate analysis).
- Delivered a 68-page technical report including full data dictionary, QA/QC checklist, and PostGIS query appendix.

GIS Analyst - Student (Capstone Project) | CIRUS Lab / SAIT, Calgary (May 2026 - Aug 2026)

Responsibilities

- Co-developed "Highway to the Trigger Zone," an avalanche trigger-zone digital twin for Morrissey Ridge, BC (72.5 km²), for the CIRUS Lab at SAIT, applying aviation/RPAS domain expertise within a 3-person team.
- Built a six-factor weighted suitability model (slope, plan curvature, aspect, ruggedness, elevation, land cover) from 1m LiDAR DEM/DSM data using an AHP weighting scheme, automated through a 3-tool ArcPy workflow.
- Led delivery of the Experience Builder digital twin, integrating 2D/3D terrain review, weather context, field validation apps (Field Maps/Survey123/QuickCapture), and an ArcGIS Dashboard into a single decision-support application.
- Applied RPAS/aviation domain knowledge to a two-run drone route planning tool, deliberately keeping final target selection with the qualified human operator rather than automating it.

Achievements

- Model identified 268 candidate trigger zones (406.6 ha) narrowed from a 72.5 km² study area, validated by transferring the unchanged workflow to an independent test area and comparing results against professional ATES terrain classifications.
- Delivered a working recorded demonstration of the two-run route planner and a print-ready mission map export requested directly by the CIRUS Lab client.

ADDITIONAL GIS PROJECTS

- Automated Python/ArcPy geoprocessing scripts - cycling-network analysis by township (Toulouse, France) and ATS/DLS land-fabric import, clip, and GPS-point location workflow (Alberta).
- Supervised multispectral land cover classification (Sibbald Lake, AB) - 84.5% overall accuracy, supporting ecological sampling design.
- Thematic cartography - choropleth, proportional symbol, and data-classification method comparison maps (GEOS 457).

EXTRACURRICULAR LEADERSHIP

President | SAIT BGIS Club (Apr 2026 - Aug 2026)

- Coordinated promotion of GIS-related events and initiatives.
- Supported communication between students, instructors, and industry partners.
- Strengthened ability to explain GIS concepts and outcomes to diverse audiences.

CERTIFICATION & MEMBERSHIP

- Certified Engineering Technologist (C.E.T.) - ASET, since May 2011
- Student Member - Alberta Society of Surveying and Mapping Technologies (ASSMT), since Feb 2026
- Small Remotely Piloted Aircraft (VLOS) Basic Operations - Transport Canada (Jul 2026)
- Integrate Data with the FME Platform - Safe Software FME Academy (Aug 2026)
- H2S Alive (Energy Safety Canada, expires May 2027) and Standard First Aid/CPR/AED Level C (Canadian Red Cross, expires Jan 2027)

PROFESSIONAL EXPERIENCE (PRE-GIS CAREER)

ILI Field Technician | United Pacific Projects, Calgary, AB (Jun 2024 - Jan 2025)

Responsibilities

- Performed infield and shop-based maintenance, repairs, and cleaning on pipeline in-line inspection tools.
- Travelled to remote sites to prepare inspection tools for surveys and advised on field conditions affecting data quality.

Electronics Repair Technician | Arcfield Canada, Calgary, AB (Jul 2019 - Apr 2024)

Responsibilities

- Tested, repaired, and maintained CF-18 avionics systems to component level using automated test equipment and approved repair schemes.
- Calibrated test equipment to OEM specifications and maintained documentation under ISO 9001/AS9100 and Airworthiness standards.

Achievements

- Maintained full calibration traceability across all serviced equipment with zero documentation non-conformances.

Wireline Electronics Technologist III | Weatherford Canada Partnership, Grande Prairie / Red Deer, AB (Jun 2009 - Mar 2019)

Responsibilities

- Diagnosed and repaired electronic and mechanical faults in wireline tools; performed preventive, corrective, and calibration maintenance.
- Participated in root cause analysis, trained new hires, and managed spare parts inventory and on-call support.

Achievements

- Recognized as a go-to technician for complex field diagnostics over a 9-year tenure, training multiple new hires on assembly and troubleshooting procedures.

EDUCATION

Bachelor of Applied Technology, Geographic Information Systems | SAIT, Calgary, AB (Jan 2026 - Present)

- Coursework in cartographic design, spatial analysis, geodatabase design, GPS field data capture, remote sensing/classification, and geoprocessing automation (Python/ArcPy, FME, ModelBuilder).

Electronics Engineering Technology Diploma | SAIT, Calgary, AB (2009)

- TAC-accredited program: circuit design and analysis, digital/analog electronics, microprocessor systems, wireless communications, and automated test and measurement; completed a capstone engineering design project.

TECHNICAL SKILLS

GIS Software

ArcGIS Pro, ArcGIS Enterprise, ArcGIS Online, QGIS, ArcGIS StoryMaps, Survey123, Experience Builder, Field Maps, QuickCapture, Dashboards

Spatial Analysis

Coordinate systems & projections, geodatabase design, spatial joins/queries, remote sensing & classification, cartographic design

Programming & Data

Python (ArcPy), SQL, PostgreSQL/PostGIS, FME, ModelBuilder, Excel

Professional & Language

Troubleshooting & root-cause analysis, technical documentation, multidisciplinary collaboration, English & Arabic (fluent), French (advanced)